TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.102
Rolling Resistance

Test Carrier:	Test Rig
Testing Facility:	Drum Test Rig of Tire Manufacturer
Status (Porsche):	19.12.2019
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(see target data sheet / ”Zieldatenblatt”)

1 Purpose

Determination of the rolling resistance coefficient on a roller dynamometer

2 Field of Application

This test method specifies the procedure by which the rolling resistance of tires is determined stationarily at several speed levels on a chassis dynamometer.

For the release test, the speed levels 30, 60, 90, 120 km/h are to be classified. In special cases, the speed levels 30, 60, 90, 120, 150, 180, 210, 240, 270 km/h can be requested from Porsche.

The procedure is based on ISO 28580. In this context, the reference tire specified in the target data sheet (current DOT: should not be older than 26 weeks) must be measured and documented for comparison.

3 Definition of Terms

Fr	= Effective tire rolling resistance force
Fl	= Bearing force (measurand)
r	= Distance wheel center-drum
R	= Drum diameter

4 Responsibility

The test is performed and documented by the tire manufacturer (self-certification). The results are made available to the PAG immediately.

Retesting:
Retesting of approval tires can be arranged by the Porsche technical department at an independent institute.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

Measured Variable	**Measurement Tolerance**	**Adjustment & Control Tolerance**
Bearing force in circumferential direction	±0.5 N	-
Drum speed	±0.2 km/h	±1 km/h
Wheel Load	±0.5% of measured variable or ±20 N	±5% of measured variable
Tire Pressure	±0.05 bar	-
Ambient Temperature	±0.3 °C	-
Tire Surface Temp.	±1 °C	-
Distance wheel center to drum surface	±1mm	-
Time	±1 s	-
Dynamic tire radius	±0.1mm	-

Measurement of tire pressure and, if not continuously measurable, tire surface temperature is recommended (not required). The ambient temperature is measured continuously.

5.1.2 Test Rig

- Drum test rig with 1.5 - 3 m diameter, reference diameter is 2 m (conversion is done via loaded tire radius)
- Drum surface: steel, smooth or safety walk
- Drum driven, speed controlled
- Test method:
  Measurement of the reaction forces at the tire rotation axis, measurement of the loaded tire radius.

5.1.3 Data Processing

N/A

5.2 Test Conditions

- Ambient temperature must be 25° C ±5° C.
  A temperature correction to the reference temperature of 25° C must be carried out according to DIN ISO 28580
- Reference temperature for air pressure setting is 20° C

5.3 Test Carrier / Test Set-Up

- Test load = wheel load according to target data sheet (no drum correction)
- Speed levels and run times according to table

Level	Runtime [min]	Speed [km/h]
1	30	30
2	20	60
3	20	90
4	20	120
5	20	150 *
6	20	180 *
7	20	210 *
8	20	240 *
9	20	270 *

*bei Anfrage

- Tire pressure according to target data sheet (free pressure build-up)
- Camber angle = 0°
- Inclined elevation angle = 0°
- The rim width, diameter and contour at the rim seat of the test rim must correspond to the respective original Porsche wheel (see target data sheet). Additional drillings, e.g. for internal temperature measurements, are permissible

5.4 Test Procedure

Pre-conditioning:
In order to improve the reproducibility of the measurement results, the new tire must undergo the following treatment before the measurement procedure: 60 minutes of tire running at 100km/h with test pressure and test load on test rim, followed by flat storage of the wheel under the temperature conditions of the test location for at least 3 hours (maximum 1 week).

Measurement procedure:
Before the measurement, the tire pressure is set when the tires are cold (20° C). The drum drives the tire under test conditions at the respective speed level with the respective time interval. The speed levels are only approached in ascending order.

After the measurement with wheel load according to the parameter table, a measurement of the internal resistances (fan resistance, bearing resistance) is carried out after each speed step by no-load measurement, whereby the drum drives the wheel with an average wheel load (50N ±5N) so that the tire just rolls along. The time span for the no-load measurement should not exceed 2 minutes (achievement of a static condition).

6 Documentation

6.1 Data Analysis

The rolling resistance shall be determined according to the state of the art. The average value of 30/60/90/120 km/h is relevant for approval.

In the result sheet, plot all measured variables and the resulting tire rolling resistance versus speed. Furthermore, the result sheet must contain the following information:

- Test date
- Tire size
- Tire make, designation and specification number
- DOT number
- Test parameters
- Name of the examiner
- Test site, test rig

6.2 Presentation of the results

In the form of the test catalog, data sheet rolling resistance F.360.102

6.3 Requirements

At least two tires per dimension shall be tested. The age of the tires should be between 4 and 28 weeks. The tire must at least reach the values specified in the target data sheet.

7 Applicable Documents

Porsche AG, data sheet: "Rolling resistance - F.360.102"

8 Distributor

EFF